

Write a complete Python script that performs numerical analysis on structural length  $L = 20.0$  meters and angle approximations for  $y = L \sin(\theta)$  using loops. The script must accomplish the following deliverables:

1. **Deliverable 1 (Numerical Tables):** Generate HTML tables comparing exact  $y$ , Maclaurin approximations for  $N = 1, 2, 3, 4$  terms, absolute errors, and percentage errors for angles  $[1^\circ, 2^\circ, 5^\circ, 10^\circ, 15^\circ, 20^\circ, 30^\circ]$ , plus a small-angle critical-angle investigation (0.1% tolerance).
2. **Deliverables 2, 3, & 4 (Matplotlib Graphs):** Generate and save three separate PNG files (deliverable2\_convergence.png, deliverable3\_comparison.png, deliverable4\_errors.png) showing:
  - Maclaurin convergence across terms  $N$  (log scale).
  - Function comparison plots (Exact vs. Maclaurin at  $0^\circ$  and Taylor at  $10^\circ$ ).
  - Absolute and percentage error comparisons for  $N = 2$  terms.
3. **Deliverable 5 (Engineering Recommendation):** Provide a structured written engineering recommendation section evaluating computational tradeoffs and accuracy.
4. Style the entire dashboard using a cohesive pastel aesthetic with light purple background (#F3E5F5), dark light purple headers (#7E57C2), soft pink accents, and soft blue highlights (#1E88E5)."